



Processing Speed Partially Mediates Executive Function Impairments in Adolescents with Congenital Heart Disease: Results from a Prospective Cohort Study

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Objective To assess processing speed, fine motor function, attention, and executive function (EF) impairments in adolescents with complex congenital heart disease (CHD) who underwent open-heart surgery during infancy.

Study design We administered a comprehensive neuropsychological test battery evaluating 5 EF domains: working memory, inhibition, cognitive flexibility, fluency, and planning and primary neurodevelopmental processes (PNPs): processing speed, fine motor function, and attention. The sample included 100 adolescents with complex CHD from a previous University Children's Hospital Zurich study, with 104 healthy controls for comparison. We generated scores for each EF domain and computed an EF summary score. Group comparisons and associations were analyzed with multiple regressions accounting for parental education. Mediation analysis explored how PNPs mediate the effect between a CHD diagnosis and EF.

Results In adolescents with complex CHD, all EF domains and the EF summary score were impaired ($\beta = 0.20$ to 0.37 , all $P < .05$). Furthermore, they exhibited slower processing speed ($\beta = 0.27$, $P < .01$) than healthy controls, with no differences in attention ($\beta = -0.07$, $P = .34$) and fine motor function ($\beta = 0.08$, $P = .34$). Processing speed showed a strong association with the EF summary score ($\beta = 0.60$, $P < .001$) and partially mediated the relationship between CHD diagnosis and the EF summary score ($\beta = 0.37$, 95% CI [0.24, 0.50], $P < .001$).

Conclusion Adolescents with complex CHD show difficulties in EFs and processing speed. Notably, processing speed is strongly associated with EFs and partly accounts for EFs disparities between patients and healthy controls. Early detection and interventions for processing speed difficulties may improve EF outcomes in these patients. (*J Pediatr* 2024;272:114091).

There is widespread agreement that children with complex congenital heart disease (CHD) are at risk for neurodevelopmental difficulties.^{1,2} In particular, executive function (EF) difficulties are parentally reported in around 12% to 17% of school-age children and adolescents with complex CHD³ and are significantly higher than the population mean.^{4,5} EFs are interrelated higher-order cognitive processes such as working memory, inhibition, cognitive flexibility, fluency, and planning.⁶ Children with complex CHD are also described as demonstrating difficulties in processing speed,^{3,7,8} fine motor function,^{7,9} and attention.^{7,10} These domains are summarized here as primary neurodevelopmental processes (PNPs). During development, difficulties in PNPs emerge at earlier stages than EF impairments.⁹ In addition, components of PNPs may influence each other, eg, fine motor and visuomotor impairments may amplify processing speed difficulties.¹¹ Similarly, in a patient group of children with ADHD, a link between attention problems and processing speed has been shown.¹² This potential cascade of effects unfolds during childhood and might impede the development of higher cognitive functions, such as EFs.¹³⁻¹⁸ However, to date, little is known about how PNPs are related to EFs in adolescents with complex CHD. One study provided evidence for an association between processing speed difficulties and EF impairments in children and adolescents with dextro-transposition of the great arteries (d-TGA).¹³ Further, a randomized controlled trial using a working memory intervention showed

CHD	Congenital Heart Disease
D-KEFS	Delis-Kaplan Executive Function System
EF	Executive Functions
PNP	Primary Neurodevelopmental Processes
RWT	Regensburger Wortflüssigkeitstest
SDS	SD Scores
SES	Socioeconomic Status
TAP	Test of Attentional Performance
WISC-IV	Wechsler Intelligence Scale for Children IV edition

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significant improvement in inhibitory control and attention in adolescents with complex CHD.¹⁹ These findings suggest a relationship between PNP and EFs in children with CHD.¹⁸ However, this needs to be firmly established and specifically tested to demonstrate the role of PNPs in the development of EFs in adolescents with complex CHD.¹⁸ Therefore, this study seeks to investigate not only EFs and PNPs in this population but also their interrelatedness.

The present study investigates hypotheses that were developed from a previously published study protocol.²⁰ We aimed to determine the spectrum and degree of impairments of EF (working memory, inhibition, cognitive flexibility, fluency, and planning) and PNP (processing speed, fine motor function, attention) in adolescents with complex CHD. Further, we investigated the role of PNPs in explaining the variability in EFs. We hypothesize that (1) adolescents with complex CHD demonstrate poorer performance than healthy peers in all EF domains and in all PNPs, (2) PNPs are positively associated with EF functions in both adolescents with complex CHD and healthy controls, and (3) PNPs mediate the impact of a complex CHD diagnosis on EFs in adolescents.

Methods

Study Design and Sample

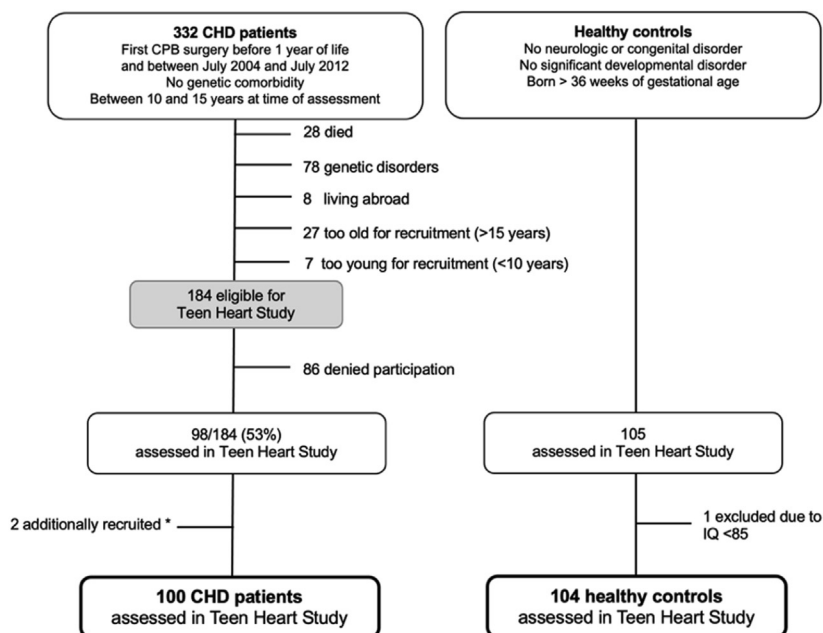
This study is part of a prospective cohort study, the Teen Heart study,²⁰ which investigated neurodevelopmental outcomes and cerebral MRI in adolescents with complex CHD and was conducted at the University Children's Hospital Zurich

between 2019 and 2021. The patients with complex CHD had undergone their first cardiopulmonary bypass surgery (CPB) within their first year of life at the University Children's Hospital Zurich between 2004 and 2012 and had not been diagnosed with a genetic comorbidity or dysmorphic syndrome prior to recruitment. At the time of assessments, they were between 10 and 15 years of age. A healthy control group was recruited among siblings and friends of the participating patients with complex CHD and through advertisements in schools and social media. Healthy controls were included if they were not diagnosed with a genetic or dysmorphic syndrome or a neurologic or significant developmental disorder (eg, learning disorder, autism spectrum disorder, and attention deficit hyperactivity disorder) and were term born (≥ 36 weeks of gestation). See **Figure 1** for recruitment procedure. The study was approved by the ethics committee of the Canton of Zurich. Written informed consent was obtained prior to participation from the participants' legal guardians and from participants if they were 14 years and older. After agreeing to participate, adolescents were invited to a neuropsychological assessment of 2–3 hours' duration to the University Children's Hospital Zurich. The assessments were performed by a psychologist or a pediatrician in training.

Measures

Executive Functions

A standardized neuropsychological test battery was used to assess EFs (**Table I**, online available at www.jpeds.com).²⁰



*Note: Patients were recruited from two longitudinal cohort studies conducted at the University Children's Hospital Zurich. a) Spillmann et al., Pediatric Research, 2021. b) von Rhein et al., JPeds, 2015. * 2 patients were additionally recruited from the developmental outpatient clinic with the same inclusion criteria but families did not originally participate in the cohort study. CHD = congenital heart disease. CPB = cardiopulmonary bypass.*

Figure 1. Recruitment flow chart.

The test battery comprised a variety of subtests, including the Digit Span forward and backward and the Letter-number sequencing of the Wechsler Intelligence Scale for Children, fourth edition,²¹ the Corsi Block Tapping Test,²² the Go/NoGo, and TAP Flexibility Test from the Test of Attentional Performance,²³ the Interference condition of the Color Word Interference Task, the Letter-Number-Switching condition from the Trail Making Task, the Filled-dots-only condition of the Design Fluency Test, and the Tower Task of the Delis-Kaplan Executive Function System,²⁴ and lastly, the s-words and animals subtests of the Regensburger Verbal Fluency Test.²⁵

An age adjustment for EFs was performed since these functions are known to strongly depend on age. Out of all EF tests, we constructed a summary score for overall EF performance. All comprehensive details on the calculation of the EF summary score, such as statistical estimates for each EF test (Supplementary Table I, online available at www.jpeds.com), importance weights and standardization constants τ (Supplementary Table II, online available at www.jpeds.com) and statistical estimates for EF domains and summary score (Supplementary Table III, online available at www.jpeds.com) have been previously published and can be found in the supplementary file.²⁶

Primary Neurodevelopmental Processes

Processing speed was evaluated through the Coding and Symbol Search subtests from the WISC-IV, and the processing speed index was calculated according to the manual.²¹ Fine motor function was assessed with the Pegboard subtest of the Zurich Neuromotor Assessment, second edition.²⁷ Attention was measured using the Alertness subtest of the Test of Attentional Performance (TAP).²³ PNPs were age-adjusted according to existing, well-established norms.

IQ

In addition to EF and PNP assessments, IQ was estimated with a corrected short version of the Wechsler Intelligence Scale for Children fourth edition (WISC-IV). This short version included 4 subtests: Matrices, Similarities, Letter Number Sequencing, and Symbol Search.^{28,29}

Participant Characteristics

Parental education, a proxy for socioeconomic status (SES), was calculated from the highest maternal and paternal education on a 6-point scale (1 = no high-school degree, 2 = high-school degree, 3 = apprenticeship, 4 = higher diploma for craftsmen or craftswomen, 5 = advanced diploma of higher education, 6 = university degree). This score ranges from 2 to 12, with higher scores indicating higher SES.³⁰ Furthermore, clinical data have been collected prospectively from the patients' charts.

Statistical Analysis

Descriptive statistics included mean and SD for the continuous variables and numbers and percentages for the categorical variables. This data were compared between groups using

Chi square and Mann–Whitney U tests. The proportion of missing values for EFs and PNPs in the sample ranged from 1% to 7%. These missing values were imputed by chained equations.³¹

Group comparisons for neurodevelopmental data, and associations between EFs and PNPs were analyzed with multiple regression analyses that adjusted for SES. Associations were analyzed with multiple regression analyses that adjusted for SES and group. Normality of residuals was checked by visual inspection of Quantil-Quantil-Plots. Multiple testing was corrected for using the false discovery rate correction method.³²

To investigate whether a CHD diagnosis exerted an effect on EFs directly and/or indirectly through PNPs, a mediation analysis was conducted. In the mediation analysis, we explored how a potential mediator (M) processing speed accounts for the association between a CHD diagnosis (predictor, X) and EFs (outcome, Y). When the mediator is introduced, the indirect effect ($a*b$), which represents the effect of X on Y while accounting for M , can be identified. The direct effect (c') quantifies the portion of X 's effect on Y which is not explained by M . The analyses were adjusted for SES and the indirect effect was obtained using a bootstrap of 5'000 resamples. Standardized beta coefficients were reported. See Figure 2 for model visualization.³³ All analyses were conducted using R version 4.1.0 statistical software.^{31,34-45} The STROBE guidelines for reporting cohort studies were followed.

Results

Sample Characteristics

Overall, 100 adolescents with complex CHD and 104 healthy controls between the age of 10 and 15 years were recruited. Notably, SES of participants was somewhat higher than in non-participants, but not statistically significant ($W = 4527$, $P = .051$). Other variables including sex ($X^2 = 0.34$, $P = .561$), length of ICU stay ($W = 4796$, $P = .106$), and a univentricular CHD ($X^2 = 0.24$, P value = .621) were not different.

Demographic and cardiac characteristics are presented in Table II. Six families filled in the questionnaires but did not attend the neuropsychological assessment. Therefore,

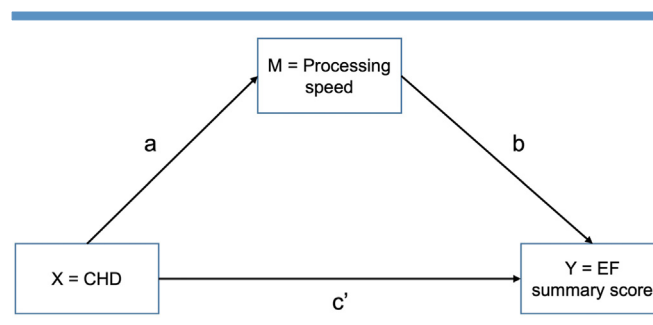


Figure 2. Mediation model visualization.

neuropsychological data were analyzed from 198 participants (95 patients, 103 healthy controls). Patients with complex CHD were significantly older ($P < .01$), and their SES was significantly lower than that of the healthy controls ($P < .01$). IQ was significantly lower in patients with complex CHD ($M = 99.2$, $SD = 14.4$) than in healthy controls ($M = 110.9$, $SD = 9.2$, $P < .001$).

Differences in EFs and PNPs Between Adolescents with Complex CHD and Healthy Controls

Using multiple linear regression models accounting for SES, we found that adolescents with complex CHD performed significantly poorer than healthy controls in the EF summary score and all EF domains ($\beta = 0.20$ to 0.37 , all $P < .05$, **Figure 3A**, **Table III**). SES was significantly associated with the EF summary score and with those for working memory, cognitive flexibility, and fluency ($\beta = 0.15$ to 0.25 , all $P < .05$). The individual subtests and comparisons between groups and with normative data are presented in Online **Table IV** available at www.jpeds.com.

For PNPs, after adjusting for SES, we found that adolescents with complex CHD performed poorer than healthy controls in processing speed ($\beta = 0.27$, $P = .003$) but not in fine motor function or in attention (**Figure 3B**, **Table III**).

Table II. Demographic and clinical characteristics stratified by group

Population characteristics	CHD (n = 95)	Control (n = 103)	P value
Innate			
Male sex	58 (61%)	48 (47%)	.058*
Age in years; Mean (SD)	13.7 (1.2)	13.0 (1.4)	.001†
Parental education; Median (IQR)	8 (6-9)	10 (8-12)	<.001†
Gestational age in weeks; Mean (SD)	39.1 (2.0)	39.2 (1.5)	.82†
Prenatal diagnosis	23 (24%)	-	
Cyanotic CHD	69 (73%)	-	
Acyanotic CHD	26 (27%)	-	
Biventricular CHD	71 (75%)	-	
Univentricular CHD	24 (25%)	-	
Preoperative (first CPB surgery)			
Preoperative saturation; Mean (SD)	83.0 (12.7)	-	
Age at surgery in months; Mean (SD)	3.4 (2.9)	-	
Intraoperative (first CPB surgery)			
Lowest temperature in °C; Mean (SD)	28.4 (6.4)	-	
Time on ECC in minutes; Mean (SD)	171.7 (72.8)	-	
Postoperative (first CPB surgery)			
Time in ICU in days; Median (IQR)	14.5 (8-23)	-	
Time in hospital in days; Median (IQR)	31.5 (20-57)	-	
Number CPB surgeries; Median (IQR)	1 (1-2)	-	

CC, extracorporeal circulation; CPB, cardiopulmonary bypass; ICU, intensive care unit; IQR, interquartile range; SD, standard deviation.

*Two-sampled Chi-squared test.

†Two-sampled, two-sided Mann-Whitney U test.

In addition, SES was significantly associated with processing speed, and attention ($\beta = -0.19$ to 0.19 , $P < .05$).

Posthoc analysis showed that there was no significant difference between patients with univentricular CHD and those with biventricular CHD in either EFs or PNPs ($\beta = -0.01$ to 0.16 , all $P > .05$).

Associations Between EFs and PNPs

Multiple linear regression examined the association between each PNP and both the EF summary score and EF domains while accounting for SES and group. The standardized beta coefficients are reported and interpreted as associations. Significant medium to large associations were found for processing speed across the EF summary score and all EF domains ($\beta = 0.29$ to 0.60 , all $P < .001$). Fine motor function only produced significant low to medium associations with the EF summary score and all EF domains ($\beta = 0.17$ to 0.34 , all $P < .05$). Attention did not show any significant associations at all (**Figure 4**).

Posthoc interaction analyses revealed interaction effects for processing speed $\times$ group on the EF summary score ($\beta = -0.17$, $B(CI-95) = -0.80(-1.45$ to $-0.16)$, $P = .01$) and processing speed $\times$ group on the EF domain cognitive flexibility ($\beta = -0.18$, $B(CI-95) = -0.90(-1.57$ to $-0.24)$, $P = .008$, **Figure 5**). The association between higher

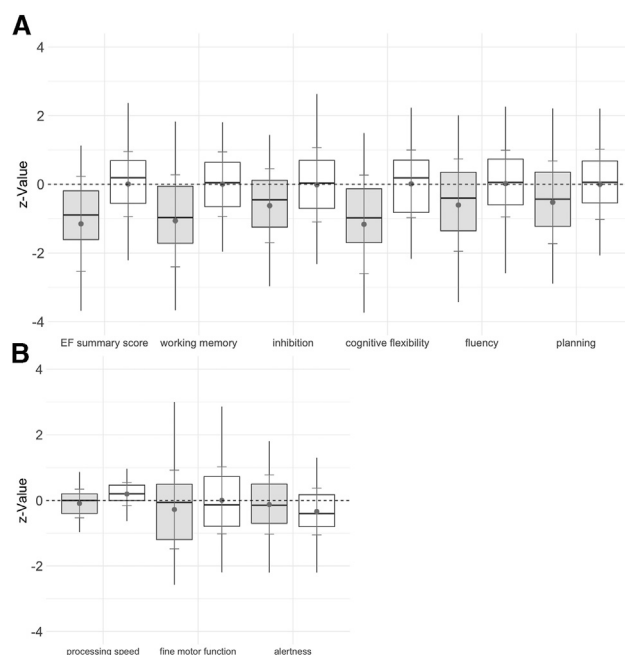


Figure 3. Differences in **(A)** executive functions between adolescents with complex CHD (light grey) and healthy controls (white) and **(B)** primary neurodevelopmental processes between adolescents with complex CHD (light grey) and healthy controls (white). The box represents the IQR. The thick, black line within the box corresponds to the median. The dashed line represents the mean of the norm population. Mean and SD are depicted in dark grey.

Table III. Results of multiple linear regression analysis of EFs and PNPs

Neurodevelopmental measures		z-values adolescents with CHD mean $\pm$ SD	z-values healthy controls mean $\pm$ SD	β [95% CI]	P value*
Executive functions	EF summary score	-1.15 $\pm$ 1.38	0.01 $\pm$ 0.95	0.37 [0.24 to 0.49]	< .001
	Group			0.19 [0.06 to 0.33]	.008
	Parental education				
	Working memory	-1.06 $\pm$ 1.33	0.00 $\pm$ 0.94	0.36 [0.24 to 0.49]	< .001
	Group			0.15 [0.02 to 0.29]	.036
	Parental education				
	Inhibition	-0.62 $\pm$ 1.07	-0.01 $\pm$ 1.08	0.24 [0.10 to 0.38]	.003
	Group			0.08 [-0.06 to 0.23]	.278
	Parental education				
	Cognitive flexibility	-1.17 $\pm$ 1.43	0.01 $\pm$ 0.99	0.33 [0.21 to 0.46]	< .001
Primary neurodevelopmental processes	Group			0.25 [0.12 to 0.38]	< .001
	Parental education				
	Fluency	-0.61 $\pm$ 1.34	0.02 $\pm$ 0.97	0.20 [0.06 to 0.34]	.010
	Group			0.15 [0.01 to 0.30]	.049
	Parental education				
	Planning	-0.52 $\pm$ 1.20	-0.00 $\pm$ 1.02	0.20 [0.06 to 0.35]	.010
	Group			0.06 [-0.09 to 0.20]	.460
	Parental education				
	Processing speed	-0.09 $\pm$ 0.44	0.19 $\pm$ 0.35	0.27 [0.13 to 0.40]	.003
	Group			0.19 [0.05 to 0.33]	.031
	Parental education				
	Fine motor function	-0.28 $\pm$ 1.20	0.00 $\pm$ 1.02	0.08 [-0.07 to 0.23]	.337
	Group			0.13 [-0.02 to 0.28]	.137
	Parental education				
	Attention	-0.13 $\pm$ 0.90	-0.34 $\pm$ 0.71	-0.07 [-0.22 to 0.08]	.337
	Group			-0.19 [-0.34 to -0.04]	.032
	Parental education				

P values are corrected for multiple testing using the FDR correction.

*Determined by multiple linear regression analysis. R^2 (EF summary score) = .220 ($n = 198$, $P < .001$); R^2 (working memory) = .191 ($P < .001$); R^2 (inhibition) = .071 ($P = < .001$); R^2 (flexibility) = .230 ($P < .001$); R^2 (fluency) = .078 ($P < .001$); R^2 (planning) = .044 ($P = .004$); R^2 (processing speed) = .138 ($P < .001$); R^2 (fine motor function) = .019 ($P = .057$); R^2 (attention) = .043 ($P = .005$).

processing speed and higher EF was stronger in adolescents with complex CHD but present in both groups ($\beta_{\text{patients}} = 0.69$, $P < .001$, $\beta_{\text{controls}} = 0.51$, $P < .001$). The same holds for the association between higher processing speed and higher cognitive flexibility ($\beta_{\text{patients}} = 0.68$, $P < .001$, $\beta_{\text{controls}} = 0.49$, $P < .001$).

Mediation of CHD Diagnosis on EFs by PNPs

In the mediation analysis, we explored how processing speed as a mediator (M) accounts for the association between a CHD diagnosis (X) and EFs (Y). We found a significant partial mediation: When processing speed was introduced as a mediator, the *direct effect* (c') quantifying the portion of the

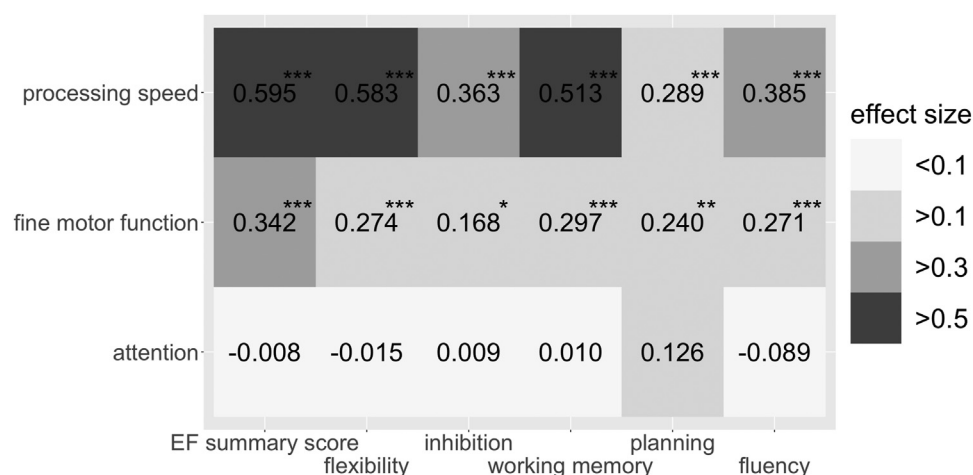


Figure 4. Heatmap of associations in the entire sample (Standardized Betas, controlled for SES and group) between primary neurodevelopmental processes and executive functions. * $P < .05$; ** $P < .01$; *** $P < .001$. Colors refer to effect sizes $r > 0.5$ = large effect, $r > 0.3$ medium effect, $r > 0.1$ small effect, $r < 0.1$ negligible effect.

effect of having a CHD diagnosis on EFs which is not explained by processing speed diminished (Figure 6).

Discussion

The current study compared adolescents with complex CHD with healthy controls in their overall EFs and performance in processing speed, fine motor function and attention, here summarized as PNPs. We found poorer performance in adolescents with complex CHD in the EF summary score and in all EF domains: working memory, inhibition, cognitive flexibility, fluency, and planning. Among the PNPs, only processing speed was impaired in the complex CHD group. Across both groups, processing speed and fine motor function were significantly associated with the EF summary score and all EF domains. Processing speed had a stronger association with both the EF summary score and the EF domain of cognitive flexibility in the complex CHD group than in healthy controls, indicating an interaction effect. Furthermore, processing speed partially mediated the effect of a CHD diagnosis on EF impairments.

The lower performances of patients with CHD in all EF domains align with previous meta-analyses,^{1,2} which showed significantly impaired EFs without a specific profile of EF impairments. Together, these findings indicate that no single EF domain is particularly affected in patients with complex CHD.

Our findings show that despite ongoing improvements in diagnosis, monitoring, and treatment of patients with complex CHD, neuropsychological sequelae persist in contemporary cohorts of such adolescents. The mechanisms underlying these impairments are not yet fully understood. Already during fetal development, altered cardiovascular physiology can lead to reduced oxygen and nutrient delivery and can thus alter brain development early on.⁴⁶⁻⁴⁹ As after birth, infants and children with complex CHD may suffer from hypoxemia^{50,51} and hypoperfusion and undergo anesthesia and cardiac surgery, these factors are also listed as potential

contributors to neurodevelopmental outcomes.⁵²⁻⁵⁴ Importantly, length of hospital stay and parental education have also been shown to predict long-term neurodevelopment.^{8,54} We also detected a significant association with a small to moderate effect size between parental education, a proxy for SES, and EFs. To guide targeted intervention decisions in clinical settings, health care providers need to focus particular attention on families with lower SES, because their children are at the highest risk for future EF impairments.

Importantly, our study extends prior research by also examining the role of PNPs, which potentially underlie EF difficulties. By doing so, we have helped identify developmental components of EF difficulties and gain insights valuable for future interventions and clinical applications.

Our study revealed slower processing speed in adolescents with complex CHD than in healthy controls but found no significant differences in fine motor function or attention. As processing speed also shows the highest association with EFs, it seems to be the PNP most critical for EFs. In the present study, processing speed was not only impaired in patients with CHD but also associated with EFs, which aligns with the few previous studies reporting on this association in children with CHD¹³ and typically developing children.⁵⁵

Consequently, we investigated whether processing speed produces a mediating effect. We found that processing speed partially mediates the effect of CHD diagnosis on EFs. Hence, our findings reinforce the idea that CHD can lead to poorer processing speed performance, which also impacts EFs.^{13,17} However, as both the variables were collected at the same timepoint, we cannot make any longitudinal assumptions in terms of developmental cascade effects, and whether the processing speed difficulties precede EF problems. This remains to be established with longitudinal data. What the mediation does show however, is the close link of processing speed and EFs, with poorer processing speed partly explaining the different EF outcomes between adolescents with CHD and healthy controls. The remaining unexplained variance must be found in other variables still.

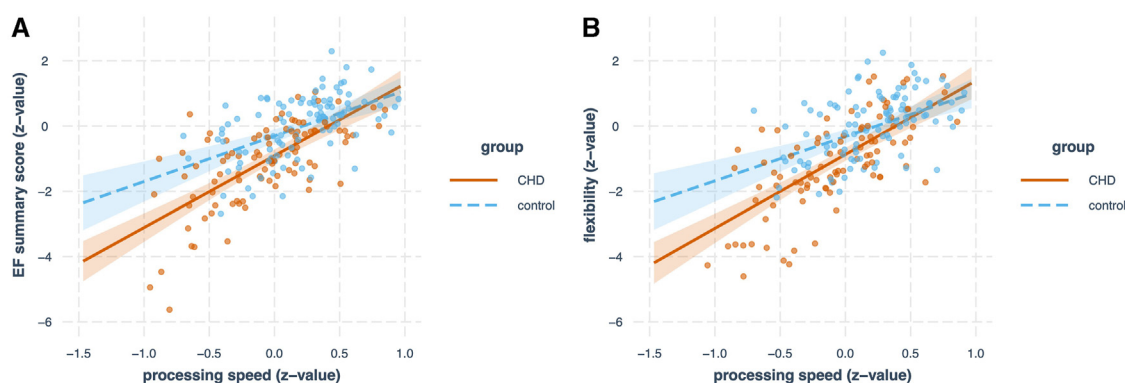


Figure 5. Associations of processing speed and the EF summary score (A) and processing speed and cognitive flexibility (B) stratified by group. Individual datapoints are represented by dots. Lines represent linear regressions for patients (orange) and controls (blue).

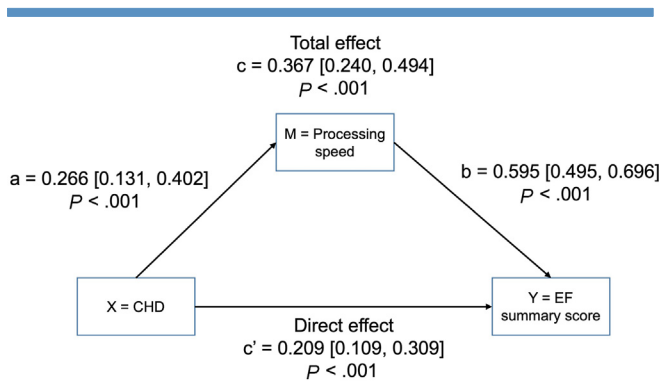


Figure 6. Depiction of mediation analysis with standardized Betas, controlled for SES, 95% CIs, and *P*-values.

This close association between processing speed and EFs has also been found in children with dextro-transposition of the great arteries,¹³ and a similar effect has been detected in a population of school-age, preterm-born children.⁵⁶ Evidence is mounting that children born very preterm and those with complex CHD share similarities in their neurodevelopmental impairments.⁵⁷

Further, the relationship between processing speed and EFs, including working memory and set-shifting, is well-documented in healthy and preterm born children.⁵⁸ The impact of processing speed on EF performance is supported by an MRI study in healthy children and adolescents.⁵⁹ In that study, better white matter maturation, measured by fractional anisotropy, was associated with improved processing speed and thus with improved reasoning ability, a higher-order cognitive function. Interestingly infants with complex CHD exhibit delayed microstructural brain development, altered neuronal network connectivity, and a higher rate of white matter injuries perioperatively.⁶⁰ Moreover, disrupted white matter microstructure integrity and reduced apparent axon density, measured by neurite orientation dispersion and density imaging have been demonstrated in previous studies in children and adolescents with complex CHD.^{61,62} In another study, altered white matter microstructure remained stable across age and led to lower functional plasticity in patients than healthy controls, underscoring how CHD constitutes a lifelong neuropsychological impairment.⁶³

When confronted with a difficult task, healthy individuals increase the activation of areas involved in an easier version of the task or recruit additional brain areas. How this occurs varies between individuals. Altered white matter microstructure may lead to less efficient use of brain networks and thus to difficulties in generating cognitive strategies in response to increased demands.^{64,65} This may explain what was observed in the patients group: a less efficient brain network not only processes information more slowly but also cannot recruit other brain areas and networks as efficiently as in healthy individuals with intact white matter microstructure. Such a mechanism would aggravate the effect of slow processing speed on higher order cognition. This explanation is

supported by the interaction effects found in the current study, in which processing speed was more strongly associated with EFs in patients than in healthy controls. Despite the limited literature on processing speed training, especially within the pediatric population, evidence suggests that it can be targeted as a distinct cognitive function.⁶⁶ Drawing on the correlation between EFs and processing speed, we advocate for cognitive interventions akin to those used for EF enhancement. For example, programs like those offered by Cognifit Inc 2022⁶⁷ or Lumosity®,⁶⁸ which feature games designed to improve both processing speed and EFs, may represent promising strategies.

We could not replicate previous findings of differences in fine motor function between adolescents with complex CHD and healthy controls.⁶⁹⁻⁷¹ However, our findings are consistent with a study in which differences were observed in fine motor functioning between patients with complex CHD and healthy controls but both groups fell within the normal range.⁷² It is possible that our sample has outgrown its fine motor function difficulties. In another study, fine motor development was significantly lower in children with cyanotic CHD before corrective surgery but improved within 2 years to nearly normal values.⁷⁰

The absence of differences in attentional performance between patients with CHD and healthy controls in the present study may be attributed to the varying attentional tests employed, as compared with prior studies where such differences were identified. Attention has diverse aspects and is assessed using various measures.⁷³ In the present study, we used a specific attentional test that measures phasic alertness.⁷⁴ Phasic alertness is the short-term focus of attention in anticipation of an event.⁷⁴ Interestingly, Hövels-Gürich et al⁷⁵ investigated multiple attentional functions in children with complex CHD, including alerting, orienting/reorienting, and executive control. They describe alerting as the achieving and maintaining of an alert state, orienting and reorienting are important when stimuli occur beyond the current attentional focus and executive control is needed to resolve conflicts between responses. Although they did not find significant differences in phasic alertness between patients and healthy controls, they observed a difference in the executive control function.⁷⁵ This finding suggests that the control of attention may be closely related to inhibition rather than achieved by a separate attentional system,⁶ which aligns with our and others' findings of inhibition difficulties in adolescents with complex CHD.¹⁹ Studies that found attentional differences between patients with complex CHD and healthy controls examined sustained attention, for example with the Test of Variables of Attention, a test for assessing ADHD and attention deficits, which is a distinct attentional construct.⁷³

This study has some limitations. First, all tests were performed cross-sectionally, which does not allow any conclusions about causal relations. Future studies should extend this work by assessing longitudinal data to determine a developmental cascade of processing speed on EFs. Furthermore, the average SES of the control group was substantially higher

than that of the patient group, signaling a noticeable recruitment bias. We are also aware, that other functions might constitute PNPs such as visuomotor integration or gross motor functions, which are frequently impaired in patients with CHD. We only included fine motor function in our assessment due to its suggested correlation with EFs.⁷⁶ Although exploring the broader implications of visuomotor and gross motor skills on EFs would be valuable, this was beyond the scope of our project and was not assessed due to time constraints. Further investigations regarding a widening of PNPs could be beneficial for future research.

In our study, there were no significant group differences in attention and fine motor performance which was unexpected. However, we only used 2 subtests to assess these functions. Attention and (fine) motor performance are complex processes which may require more comprehensive testing (eg, sustained and divided attention, vigilance).²³ Future studies should assess PNPs in more detail and relate them to EFs – preferably in a longitudinal design – to advance our understanding of developmental cascading effects. In addition, we did not perform a structured clinical assessment of ADHD and thus cannot report on the prevalence of ADHD in our cohort.

Nevertheless, this study exhibits multiple strengths. The study sample is large and well powered. Further, by addressing not only EFs but also PNPs, we were able to explore their relationship comprehensively and to contribute to the understanding of EF impairments in adolescents with complex CHD. We also used a large number of neuropsychological tests, which allow diverse EF domains to be comprehensively captured.

While previous studies have used summary scores of EF by averaging test scores,⁷⁷⁻⁸⁰ we used an advanced approach to account for data distribution and that different test measures had different variances, as well as to adequately adjust for age. Importantly, this EF summary score correlates with hormonal stress markers²⁶ and structural brain networks,⁸¹ underscoring its conceptual relevance.

In conclusion, our study revealed that adolescents with complex CHD show impaired EFs and processing speed. Notably, a strong association was found between processing speed and EFs, and this association was stronger in patients than in healthy controls. Further, processing speed mediated the impact of CHD diagnosis on EF impairments. The clinical implications of our findings are that an early evaluation of processing speed deficits and a timely therapeutic intervention, potentially through computerized paradigms, might enhance performance in this domain and subsequently have a protective effect on future EF. ■

CRedit authorship contribution statement

Alenka S. Schmid: Writing – original draft, Methodology, Formal analysis, Data curation. **Melanie Ehrler:** Writing – review & editing, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Nadja**

Naef: Validation, Investigation. **Oliver Kretschmar:** Methodology, Conceptualization. **Valentin Rousson:** Methodology, Formal analysis. **Ruth Tuura:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Flavia M. Wehrle:** Writing – review & editing, Methodology, Conceptualization. **Beatrice Latal:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare no conflicts of interest.

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